

PROJECT 3 · DISTRIBUTED AI FOR HEARING DEVICES

Edge-Cloud Audio Processing Framework

A software prototype: hearing aid → smartphone → cloud — lightweight edge features, deep on-device inference, cloud model updates.

Feature extract → WavLM → classify → cloud update

MOTIVATION

Why split across three devices?

Hearing aids have strict power and compute limits. Phones have more capacity. The cloud can improve models over time.



Hearing aid

Tiny battery, DSP only — extract compact audio features, not full neural nets.



Smartphone

Runs the heavy encoder (WavLM) and a small classifier head near the user.

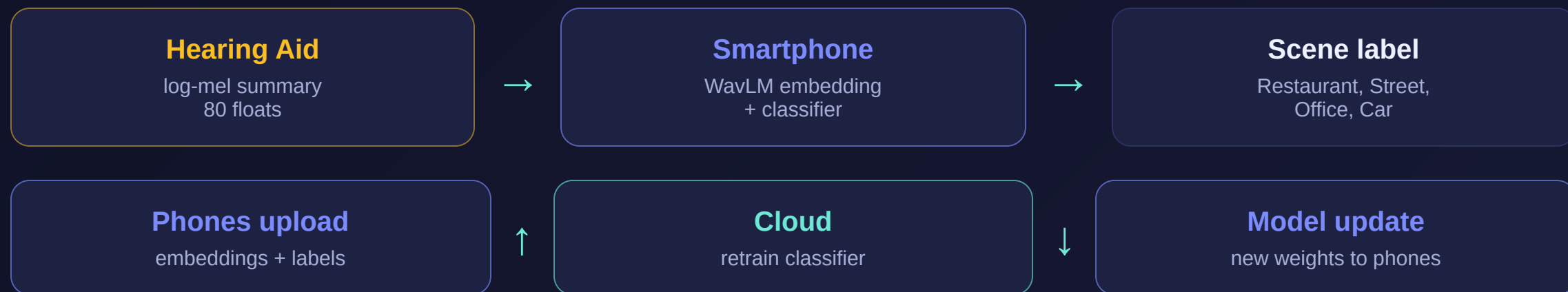


Cloud

Aggregates feedback from many users and pushes an updated classifier back.

ARCHITECTURE

Three-tier pipeline



Lightweight feature extraction

No neural network on the device. We compute a log-mel spectrogram and send a tiny summary over Bluetooth.

What it computes

- 40 mel bands × mean energy
- 40 mel bands × standard deviation
- Total: **80 floats** per clip

Why it matters

- Raw 10 s audio ≈ 160 000 samples
- Edge packet ≈ **2000× smaller**
- Good for telemetry & bandwidth checks

Deep model inference with WavLM

WHAT	DETAILS
Encoder	<code>microsoft/wavlm-base-plus</code> (frozen, from Hugging Face)
Input	16 kHz mono waveform (up to 10 s)
Embedding	Mean-pool frame outputs → 768-dim vector
Head	Logistic regression (trainable on phone, updatable from cloud)

The expensive encoder runs once per clip; only the tiny classifier head changes when the cloud pushes an update.

Aggregate feedback and push updates

1. Each phone only experiences **some** scenes, so alone it has blind spots.
2. Phones upload compact (**embedding, label**) pairs — never raw audio.
3. Cloud merges the partial views (demo: two phones) and retrain the shared head.
4. The stronger classifier is pushed back to every phone (version bump).

Prototype only: production would add encryption, federated learning, and signed OTA updates.

DEMO TASK

Acoustic scene classification

Predict the environment around the wearer so the hearing aid can switch noise-reduction profiles.



Restaurant

cafe chatter



Street

traffic, crowds



Office

keyboards, talk



Car

engine, road

Model and dataset

Model

`microsoft/wavlm-base-plus`

Loaded with `transformers`. Frozen encoder for embeddings.

Dataset

`Mahia/TUT2017`

TUT Acoustic Scenes 2017 mirror. Demo downloads ~20 clips per scene (~80 total).

Label mapping: Restaurant ← `cafe/restaurant`, Street ← `city_center`, Office ← `office`, Car ← `car`

Ten clear steps

1

Install deps

2

Configure

3

Load HF data

4

Hearing-aid
features

5

Load WavLM

6

Extract
embeddings

7

Phone classifier

8

Cloud retrain

9

Apply update

10

End-to-end demo

Notebook: `edge_cloud_audio_processing.ipynb`

RESULTS

Cloud aggregation fixes each phone's blind spots

One phone alone

50%

only saw 2 of 4 scenes

After cloud update

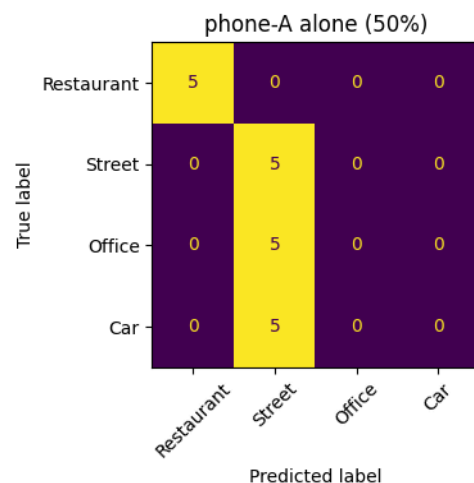
100%

all 4 scenes covered

Edge bandwidth

2000×

packet vs raw audio



Left: phone-A alone fails on scenes it never recorded. Right: the cloud-updated head gets every scene.

One clip through the full pipeline

```
wav → hearing_aid_features(wav)    # 80-float edge packet  
wav → smartphone_embed(wav)        # 768-dim WavLM vector  
emb → phone_clf.predict(emb)       # scene + confidence
```

True scene : Restaurant

Predicted : Restaurant

Edge packet: 80 floats

If the prediction is wrong, the phone would upload the embedding; the cloud includes it in the next update cycle.

TAKEAWAYS

What this prototype shows

- **Edge-cloud split** lets hearing aids stay low-power while phones run deep models.
- Devices see **partial views**; the cloud combines them into a model better than any phone alone (50% → 100%).
- The **cloud only updates the small head** — cheap to retrain and easy to deploy.
- Both **model and dataset** come from the Hugging Face Hub.

EXTENSIONS

How to go further

Speech enhancement

Replace scene classification with a denoising model (e.g. MetricGAN+) on the phone tier.

Federated learning

Train on-device without uploading raw audio — only gradients or embeddings.

Real DSP

Port the hearing-aid front-end to an embedded DSP chip instead of librosa.

More scenes

Use the full TUT2017 dataset (15 classes) for a richer environment model.

THANK YOU

Questions?

Notebook: `edge_cloud_audio_processing.ipynb`

Slides: `edge-cloud-audio-processing.pdf`

Model: `microsoft/wavlm-base-plus` · Data: `MahiA/TUT2017`

[Run the notebook top to bottom](#)